

outlets of category 13A in the same room be connected to the same cut breaker in the same DB. However, this is not the case in non-domestic premises. All socket outlets shall not be less than 2m away from one another horizontally.

Busbars: A busbar in domestic distribution refers to thick strips of copper/Aluminium that conduct electricity within a switchboard, DB or other electrical apparatus. It can also be found in substations inside feeder pillars. Each busbar has a current carrying capacity that is determined by its size and the type of conducting material employed. Busbars are typically flat strips or hollow tubes, ~~these~~ shapes are adopted because they allow heat to dissipate more efficiently due to the high surface area to cross-sectional area ratio.

DB and Consumer Unit:

A DB is a mounting enclosure for multiple electrical cut breakers, these are generally placed in columns and aligned ^{with} small single & boxes with cut breaker in just one row (and three rows for 3-Ø). DB are ^{typically} ~~usually~~ found in central locations ^{within} ~~outside~~ buildings as much as possible and often serve as the point at which electricity is distributed within a building. Cut breakers can be used to manually de-energize electrical cuts when the downstream wiring is being serviced.

Page-8 Domestic Internal distribution

Fig. 5.0 shows a 3- ϕ , 5-way distribution board containing 15 mini cut breakers. The supply to the DB passes through ELCB to the RCD switch then to the ~~residual~~ busbars, one for each phase and a separate neutral conductor, each busbar contains 5 MCB of different current ratings such as 5A, 10A, 15A, 20A, 25A, 30A etc.

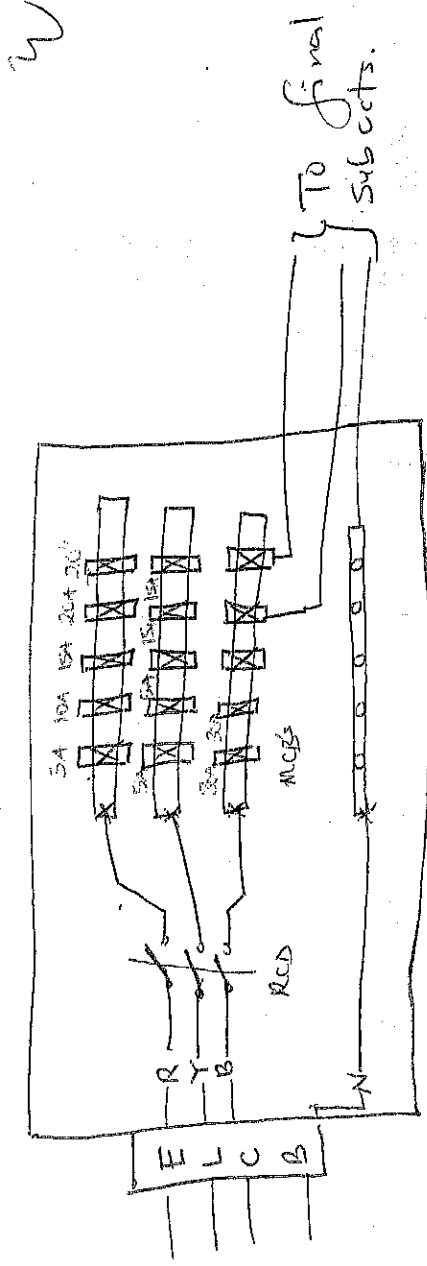


Fig 5.0: 3- ϕ , 5-way DB

The consumer's isolator and the DB can be combined to form an integrated control unit known as consumer control unit, for a small domestic premises, the ELCB, meter, isolator (cut out fuse and/or change over gear) and distribution board usually in close proximity and accessible, constitutes the consumer's switchgear which needs to be protected from improper handling.

A single ϕ 240V supply is usually taken from ϕ and N to final subccts such as fan, switches, lighting etc. ~~point, water heaters and pumps, A/c's etc.~~ In domestic installation, all socket outlets in any room should be connected to ~~the same~~ ϕ and socket

PVC Insulated and Sheathed Cables: This type of cable is used in commercial and domestic installations, which may be clipped direct to a surface, sunk in plaster or installed in conduit or trunking. It is the simplest and the least expensive. The conductors are covered with a colour-coded PVC insulator and then contained singly or with others in PVC outer sheath.

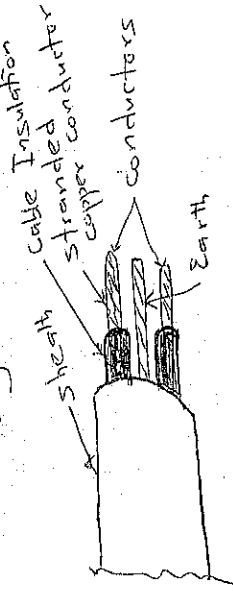


Fig. 7.0(a): A twin and earth PVC insulated and sheathed cable

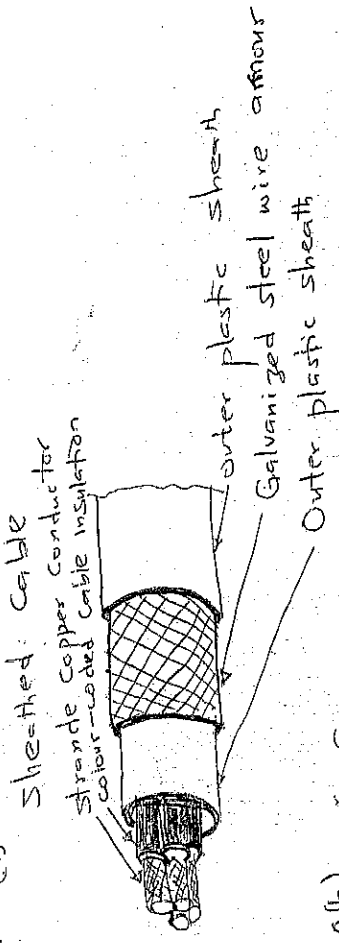


Fig. 7.0(b): A four-core PVC/SWA cable

PVC/SWA Cable: PVC insulated steel wire armoured cable are used for wirings under ground cables btw buildings, for mains supplies to dwellings, rising sub-mains and industrial installations. They are used where some mechanical protection of the cable is required. The galvanized steel wire armour provides the mechanical protection and the outer ^{plastic} sheath protects them from corrosion.

Mineral Insulated (MI) Cable: This type of cable has a seamless copper sheath which makes it water proof and fire and corrosion-resistant. This type makes it the only cable choice for hazardous or high-temp installations such as oil refineries and mechanical

works, boiler houses and furnaces, petrol pump and fire alarm installations.

The cable has a small overall diameter in comparison to alternative cables and may be supplied as bare copper or with a PVC oversheath. The Cu conductors are embedded in white powder, magnesium oxide, which is non-aging and non-combustible, but ~~which~~ ^{it is} hygroscopic (which means that it readily absorbs moisture from the surrounding air, unless adequately terminated).

~~Other cables include:~~

- Low Smoke and Fume Cables (LSF). This gives very low smoke and fumes if they are burned in a burning building.

- FP 200 Cable. This is similar to MI Cable and having a circular tube shape (or the shape of the pencil). However, it is much simpler to use, and terminate than an MI cable. The cable is a fire resistant and primarily intended for use in fire alarms and emergency lighting installations or it may be embedded in plaster.

Cable Installation and Wiring Systems

When installing cables for the purpose of wiring systems such as ~~Cables~~ Conduit, trunking or Cable tray. the cables must be installed horizontally or vertically and fixed securely to the fabric of the building.

1. Conduit installation: A conduit is a tube, channel or pipe in which insulated conductors are enclosed/contained. The conduit may serve/replace the PVC outer sheath of cable thereby providing mechanical

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protection for the cables. Conduit wiring is very popular in ~~both~~ commercial, domestic and industrial applications. There are three ^{basic} types of conduit used in electrical installations which include Steel, PVC and flexible.

Cable Parameters

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To be discussed in class

- (i) Length of run
- (ii) Ambient temp
- (iii) Grouping of cts.
- (iv) Disposition of cables
- (v) Type of sheathing
- (vi) Rating $\begin{matrix} \text{supply} & \text{distribution} \\ \text{VV} = 2.5\% & \text{VV} = 6\% \end{matrix}$
- (vii) Voltage drop $\begin{matrix} \text{supply} & \text{distribution} \\ \text{VV} = 2.5\% & \text{VV} = 6\% \end{matrix}$
- (viii) Class of excess current protection (2.5% of harmonics, 6% of fluctuation/variation)

VD = voltage drop
VV = voltage variation

Cable Selection Parameters

Parameters for selection and utilization of cable system for wiring a building differs from that of design and construction of cable. In selection and utilization of cables, certain relevant factors need to be considered, probably the most significant factor in cable selection and installation are the eight factors listed above.

All cables have certain resistance which give rise to heat when current passes through them, and the size, and the material, load connected and temp determines its withstanding ^{of conductor} capabilities of the cable.

BS6004 PVC insulated cables are suitable for operation as long as the temp of the conductor doesn't exceed 70°C.

whereas M.I. Cable can operate at higher temp of up to 135°C . If the environment ~~of the~~ in which the cable is to be installed is subject to extreme heat/temp/fire or other abnormal hazards, ^{then} consideration should be given to the selection of cables with appropriate insulation and sheathing materials.

Single pvc insulated conductors are usually drawn in to the installed conduit to complete the installation. Having decided upon the type, size and number of cables required for a final circuit, it is then necessary to select the appropriate size of conduit to accommodate those cables. The tables in Appendix 5 of the On site Guide describe a "factor system" for determining the size of ~~cable~~ ^{conduit} required to enclose number of conductors as seen in table 1(a) and 1(b) given below. the method is as follows

- i. Identify the cable factor for the particular size of conductor (from table 1(a)).
- ii. Multiply the cable factor by the number of conductor (with the same size) to give sum of cable factors
- iii. Identify the appropriate part of the conductor factor given by the length of run, and number of bends (table 1(b)).
- iv. The correct size of ~~conduit~~ ^{conduit} to accommodate the cable is that is the conduit size which has size that is equal to or greater than the ~~cable~~ ^{sum of} cable factor, but ^{the closest} size should be, ~~preferably~~ ^{preferably} be selected to minimize space as well as cost.

Example

Four 1.5mm² PVC insulated cables are to be drawn into a plastic conduit which is 5m long between boxes and contains two bends. A 4mm² PVC insulated CPC is also included. Determine the ^{optimum} appropriate size of conduit to contain these conductors.

The factor for one 1.5mm^2 Cable = 22

The factor for one 4mm^2 Cable = 43

The Sum of cable factors

$$= (22 \times 4) + (43 \times 1) = 131$$

Considering 5m length of run and ~~two~~ bends, the smallest Conduit with a cable factor greater than or equal to 131 is the ~~16mm~~ ^{20mm} Conduit having a cable factor of

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Trunking Installation

A trunking is an enclosure provided for the protection of cables which usually has square or rectangular cross-section, having one removable side. Trunking may be thought as a more accessible conduit size and for industrial and commercial installers, it is replacing the larger ^{conduit} sizes.

Trunking is supplied in 3m length and varies in cross-sections ~~are~~ measured in mm from 50 x 50 up to 300 x 150. Most trunking is available in either steel or plastic e.g.

1. Metallic trunking made from mild steel sheet, coated with grey or silver enamel paint.
2. Non-metallic trunking; this type of trunking and its accessories are also available in high-impact PVC, the accessories are usually secured to the length of trunking with a PVC solvent adhesive.
3. Mini-Trunking. This is a very small PVC trunking, ideal for surface wiring in domestic and commercial installations such as offices. The trunking has a cross section of $16 \times 16\text{mm}$, $25 \times 16\text{mm}$, $35 \times 16\text{mm}$ or $35 \times 25\text{mm}$.

and is ideal for switch drops or for housing auxiliary cts such as telephone or audio equipment wiring

Skirting trunking: A trunk manufactured from PVC or steel and in the shape of a skirting board is frequently used in commercial buildings such as hospitals, labs and offices. The trunking is fitted around the walls of a room and contains wiring for socket outlets and telephone pts which are mounted on the lids.